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AIM:Demo of OpenAI/TensorFlow Tools
CODE:

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# ============================================================
# TRAILHEAD CAREER GUIDANCE CHATBOT USING TENSORFLOW
# ============================================================

import tensorflow as tf
import numpy as np

print("TensorFlow Version:", tf.__version__)

# 1. Training Data

texts = [
    "hello", "hi", "hey", "good morning",
    "who are you", "what is your name", "what can you do",
    "what is trailhead",

    "what is cybersecurity",
    "what does a cybersecurity analyst do",
    "how does a cybersecurity analyst work",
    "how to become a cybersecurity analyst",
    "what skills are needed for cybersecurity",

    "what is software development",
    "what does a software developer do",
    "how does a software developer work",
    "how to become a software developer",
    "what skills are needed for software development",

    "what is data science",
    "what does a data scientist do",
    "how does a data scientist work",
    "how to become a data scientist",

    "what is web development",
    "what does a web developer do",
    "how does a web developer work",
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    "how to become a web developer",

    "what is it support",
    "what does application support do",
    "how does technical support work",

    "what skills should I learn",
    "which career should I choose",
    "what career options are available",
    "how to prepare for an interview",
    "how to make a resume",
    "which courses should I take",

    "thank you", "thanks",
    "bye", "goodbye", "exit"
]

# Intent labels
# 0 Greeting | 1 About | 2 Cybersecurity
# 3 Software | 4 Data Science | 5 Web Development
# 6 IT Support | 7 Skills | 8 Career
# 9 Interview | 10 Resume | 11 Courses
# 12 Thanks | 13 Goodbye

labels = [
    0, 0, 0, 0,
    1, 1, 1, 1,

    2, 2, 2, 2, 2,
    3, 3, 3, 3, 3,
    4, 4, 4, 4,
    5, 5, 5, 5,
    6, 6, 6,
    7, 8, 8, 9, 10, 11,
    12, 12,
    13, 13, 13
]

# 2. Convert Text into Numerical Features
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vectorizer = tf.keras.layers.TextVectorization(
    output_mode="tf_idf"
)

vectorizer.adapt(texts)

X = vectorizer(np.array(texts))

# 3. Create Neural Network

model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(X.shape[1],)),
    tf.keras.layers.Dense(32, activation="relu"),
    tf.keras.layers.Dense(16, activation="relu"),
    tf.keras.layers.Dense(14, activation="softmax")
])

# 4. Compile Model

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

# 5. Train Model

model.fit(
    X,
    np.array(labels),
    epochs=100,
    verbose=0
)

print("Chatbot training completed!")

# 6. Responses
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responses = {
    0: "Hello! Welcome to Trailhead. How can I help you?",
    1: "I am the Trailhead Career Assistant. I help students explore
careers, understand job roles, and discover the skills needed for their
future.",

    2: """Cybersecurity Analyst

What is the job?
A cybersecurity analyst protects computers, networks, and data from cyber
threats.

How does the work happen?
1. Monitor systems and security alerts.
2. Investigate suspicious activities.
3. Identify and contain threats.
4. Help fix security weaknesses.
5. Document incidents and improve protection.

Skills: Networking, Linux, Python, and security fundamentals.

Career path: Learn networking, practise in safe security labs, build
projects, and prepare for entry-level security roles.""",

    3: """Software Developer

What is the job?
A software developer creates and maintains computer programs and
applications.

How does the work happen?
1. Understand user requirements.
2. Plan the application.
3. Write and test code.
4. Find and fix errors.
5. Release and maintain the software.

Skills: Python or Java, problem-solving, SQL, Git, and testing.
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Career path: Learn programming, practise coding problems, build projects,
and prepare for technical interviews.""",

    4: """Data Scientist

What is the job?
A data scientist analyses data to discover patterns and support decisions.

How does the work happen?
1. Collect relevant data.
2. Clean missing or incorrect values.
3. Analyse data using statistics and charts.
4. Build and evaluate machine-learning models.
5. Explain the results.

Skills: Python, SQL, statistics, pandas, and machine learning.

Career path: Learn Python and statistics, analyse datasets, build ML
projects, and create a portfolio.""",

    5: """Web Developer

What is the job?
A web developer creates and maintains websites and web applications.

How does the work happen?
1. Plan the website.
2. Design pages using HTML and CSS.
3. Add functionality using JavaScript.
4. Connect databases and backend services.
5. Test and publish the website.

Skills: HTML, CSS, JavaScript, databases, and Git.

Career path: Learn web fundamentals, create websites, practise backend
development, and publish projects.""",

    6: """IT / Application Support
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What is the job?
IT support professionals troubleshoot application, database, and system
problems.

How does the work happen?
1. Receive a reported issue.
2. Examine error messages and logs.
3. Troubleshoot the problem.
4. Fix it or escalate it to the appropriate team.
5. Document the solution.

Skills: SQL, troubleshooting, operating systems, networking, and
communication.

Career path: Learn SQL, practise troubleshooting, understand application
logs, and prepare for support interviews.""",

    7: "Important skills include programming, SQL, problem-solving,
communication, teamwork, Git, and project development. Choose additional
skills according to your career goal.",

    8: "You can explore Software Development, Cybersecurity, Data Science,
Web Development, Cloud Computing, and IT Support. Choose a career
according to your interests, strengths, and learning goals.",

    9: "Prepare your introduction, revise technical concepts, practise
explaining your projects, and prepare common HR questions. Mock interviews
can help you build confidence.",

    10: "A fresher's resume should include education, technical skills,
projects, certifications, achievements, and relevant college activities.
Keep it clear, honest, and tailored to the job.",

    11: "You can explore freeCodeCamp, Harvard CS50, Microsoft Learn, and
Cisco Networking Academy. Choose courses relevant to your career goal.",

    12: "You're welcome! Keep learning and exploring with Trailhead.",

    13: "Goodbye! Best wishes for your career journey!"
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}

# 7. Start Chatbot

print("\n========================================")
print("     TRAILHEAD CAREER CHATBOT")
print("========================================")
print("Ask about careers, jobs, and required skills.")
print("Type 'exit' to stop.\n")

while True:

    message = input("You: ").strip()

    if message.lower() in ["exit", "quit", "bye", "goodbye"]:
        print("Bot:", responses[13])
        break

    if not message:
        print("Bot: Please enter a question.")
        continue

    user_input = vectorizer(np.array([message]))

    prediction = model.predict(user_input, verbose=0)

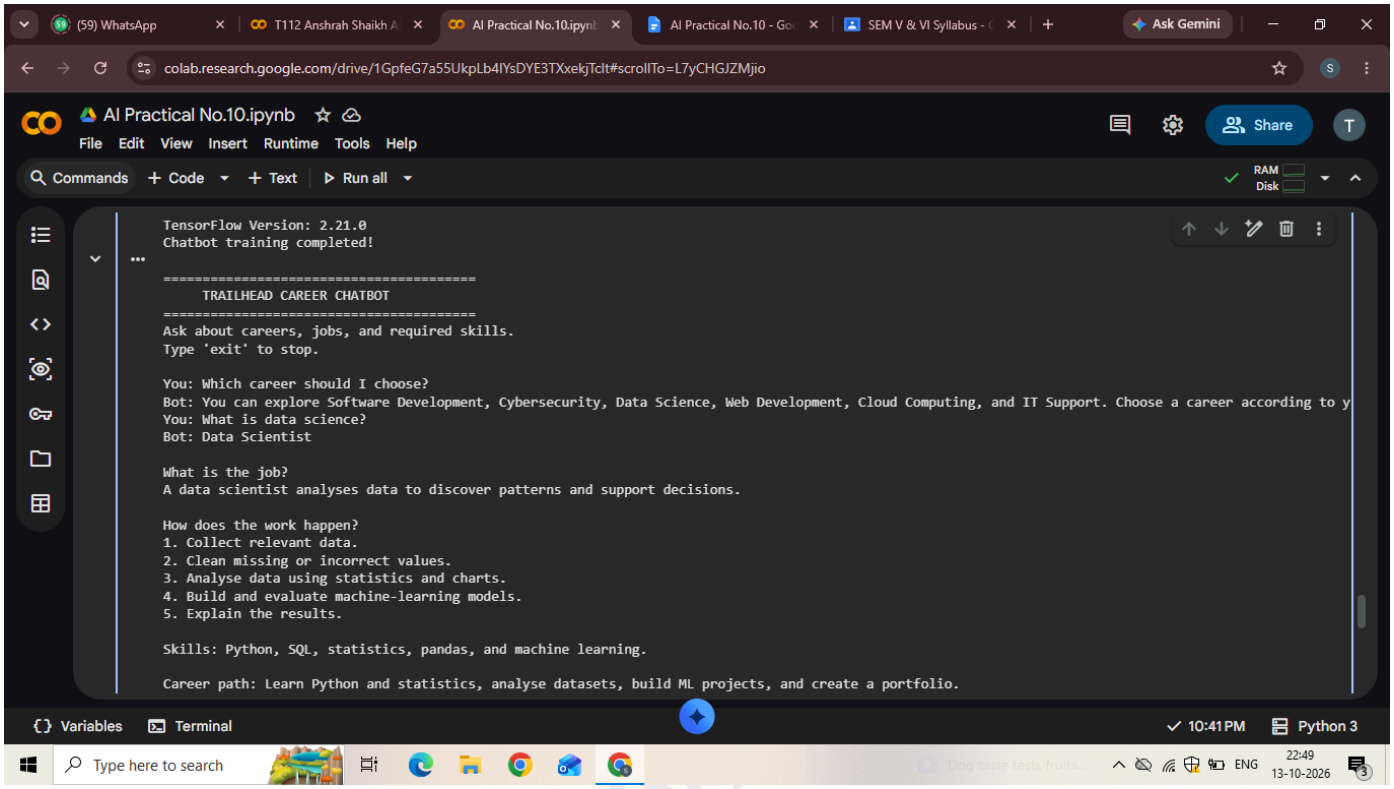
    intent = np.argmax(prediction)
    confidence = np.max(prediction)

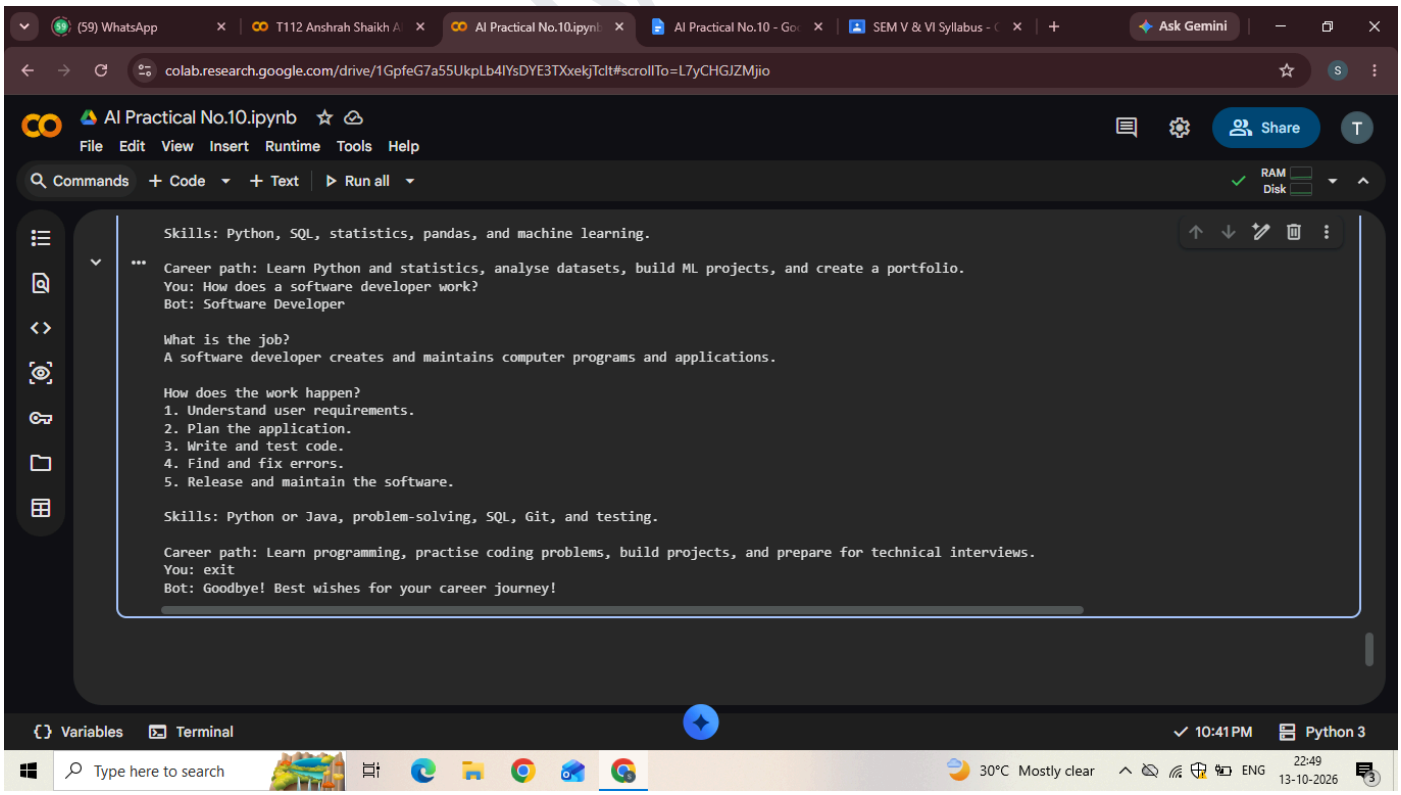
    # Handle questions the chatbot may not understand
    if confidence < 0.50:
        print(
            "Bot: I couldn't understand that question. "
            "Try asking about cybersecurity, software development, "
            "data science, web development, IT support, skills, "
            "courses, resumes, or interviews."
        )
    else:
        print("Bot:", responses[intent])
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OUTPUT:





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